Model-Informed Machine Learning for Macro-Finance Forecasting

Embedding Real Business Cycle Structure into Neural Network Priors with Empirical Fine-Tuning

Introduction and Motivation

Teaching Stochastic Economics to a Neural Network

- Macroeconomic models and machine learning solve different problems.
  - Structural models are interpretable and theory-consistent, but restrictive and often misspecified.
  - Machine learning gives flexibility and predictive capacities, but lacks economic structure.
- Can we combine the two to improve prediction of macroeconomic states, and as a proxy, extract relevant asset price signals or policy path?

Approach

- Derive and calibrate a structural macro model, and use it to generate economically consistent synthetic data.
- Train a neural network in a two-stage approach:
  - Pretraining: simulate and feed in large datasets generated by the calibrated model.
  - Fine-tuning: adapt the network using real macroeconomic and financial data.

Outputs

- Estimated latent states.
- Improved forecasts of macro variables and asset-relevant signals.

Benchmark

- Compare against traditional statistical models and baseline.
- Evaluate each phase of the network against others.

Agenda

1. Model and Calibration
2. Synthetic Pretraining
3. Fine Tune Stage 1
4. Fine Tune Stage 2
5. Results and Evaluation
6. Discussion

Model and Calibration

Model Derivation

Real Business Cycle Model with Rational Expectations.

Households are Utility Maximizers, and Choose
- Consumption (Improves Utility).
- Labor Supply (Harms Utility).
- Next Period Capital.

Firms and Profit Maximizers, and Choose
- Capital.
- Labor.
- Produce on a Cobb-Douglas Function.

Productivity and Capital/Investment Shocks are AR(1)

Market Clearing Conditions
- Resource Constraint.
- Capital Evolution.

Variable	Definition
$\widehat{c}_t$	Consumption
$\widehat{k}_t$	Capital Stock
$\widehat{n}_t$	Labour Supply
$\widehat{z}_t$	Productivity Shock
$\widehat{\xi}_t$	Capital/Investment Shock
$\widehat{i}_t$	Investment
$\widehat{y}_t$	Output
$\widehat{r}_t$	Return to Capital

Model Derivation (Continued)

Solve Household and Firm Problems

- Euler and Labour Supply Conditions.
- Wage and Rental Rate Conditions.

Solve Steady States as Functions of Parameters.

Log Linearize

- For any variable a, set $\widehat{a}_t = \log a_t - \log a$
- Then, $\widehat{a}_t$ represents log deviations from the steady state.

Parameter	Definition
α	Capital Share
β	Discount Rate
δ	Depreciation Rate
ϕ	Disutility of Labour
ρ_z, ρ_ξ	Shock Persistence
σ_z, σ_ξ	Shock Volatility

Model Derivation (Continued) - Derived Linearized Model Equations

Euler Equation

$$\widehat{c}_t = E[\widehat{c_{t+1}}] - \beta r E[\widehat{r_{t+1}}]$$

Rental Rate of Capital

$$\widehat{r}_t = \widehat{z}_t + (\alpha - 1)\widehat{k}_t + (1 - \alpha)\widehat{n}_t$$

Labour Market Equilibrium

$$(\phi + \alpha)\widehat{n}_t + \widehat{c}_t = \widehat{z}_t + \alpha\widehat{k}_t$$

Production Function

$$\widehat{y}_t = \widehat{z}_t + \alpha\widehat{k}_t + (1 - \alpha)\widehat{n}_t$$

Capital Accumulation

$$\widehat{k_{t+1}} = (1 - \delta)\widehat{k}_t + \delta\widehat{i}_t + \delta\xi_t$$

Capital/Investment Shock AR(1) Process

$$\widehat{\xi}_t = \rho_\xi\widehat{\xi_{t-1}} + \sigma_\xi\varepsilon_t^\xi$$

TFP Shock AR(1) Process

$$\widehat{z}_t = \rho_z\widehat{z_{t-1}} + \sigma_z\varepsilon_t^z$$

Resource Constraint

$$\widehat{y}_t = \frac{c}{y}\widehat{c}_t + \frac{i}{y}\widehat{i}_t$$

Model Derivation (Continued)

Reduce System to Euler Equation, Capital Evolution, and Shocks.

Rewrite as Matrix System.

$$AE_t[s_{t+1}] = Bs_t + C\epsilon_{t+1}$$

Solve Using Generalized Eigenvalues.

- Seperate Stable and Unstable Roots, and Keep Stable Solution with:

$$\max |\lambda(P)| < 1$$

Obtain State Space.

$$s_{t+1} = Ps_t + Q\epsilon_{t+1} \text{ where } s_t = \begin{bmatrix} \widehat{c}_t \\ \widehat{k}_t \\ \widehat{z}_t \\ \widehat{\xi}_t \end{bmatrix}, \qquad \varepsilon_t = \begin{bmatrix} \varepsilon_t^z \\ \varepsilon_t^\xi \end{bmatrix}$$

Above, P is the Propagation of State Variables, and Q is the Impact of Shocks on the System.

Consumption Function $\hat{c}_t = a_k \hat{k}_t + a_z \hat{z}_t + a_\xi \hat{\xi}_t$

Model Calibration via Simulated Method of Moments

Simulate Model Given Parameters → Compute Model Moments → Match to Empirical Moments

Fix Parameters

- Alpha (Capital Share) = 0.33
- Beta (Discount Rate) = 0.99
- Delta (Depreciation Rate) = 0.025

Parameters to Estimate are $\theta = (\phi, \rho_z, \rho_\xi, \sigma_z, \sigma_\xi)$

Data from FRED for Real GDP, Consumption, Investment

- GDPC1, PCECC96, GPDIC1 respectively.

Construct Quarterly Log Deviation Series $g_t^y = \Delta \log y_t, \quad g_t^c = \Delta \log c_t, \quad g_t^i = \Delta \log i_t$

HP Filter log GDP to isolate business cycle fluctuations.

Model Calibration via Simulated Method of Moments (Continued)

Target Moments

$$m^{data} = \begin{bmatrix} E[c/y] \\ \mathrm{std}(g^y) \\ \mathrm{std}(g^c) \\ \mathrm{std}(g^i) \\ \frac{\mathrm{std}(g^c)}{\mathrm{std}(g^y)} \\ \frac{\mathrm{std}(g^i)}{\mathrm{std}(g^y)} \\ \mathrm{acf}(y\text{-cycle}) \\ \mathrm{acf}(g^i) \end{bmatrix}$$

Optimization Problem*

$$\hat{\theta} = \arg\min_{\theta} (m^{model}(\theta) - m^{data})' W (m^{model}(\theta) - m^{data})$$

*With higher weights on persistence and investment volatility.

Model Parameter Estimates via Simulated Method of Moments

Parameter	Value
α	0.3300
ϕ	1.2016
ρ_z	0.7285
ρ_ξ	0.8558
σ_z	0.0057
σ_ξ	0.1000
β	0.9900
δ	0.0250
c/y	0.6453

Moment	Data	Model	Difference
$E[c/y]$	0.6435	0.6435	0.0000
$\mathrm{std}(g^y)$	0.0105	0.0094	-0.0011
$\mathrm{std}(g^c)$	0.0103	0.0089	-0.0013
$\mathrm{std}(g^i)$	0.0405	0.0363	-0.0042
$\mathrm{std}(g^c)/\mathrm{std}(g^y)$	0.9760	0.9484	-0.0276
$\mathrm{std}(g^i)/\mathrm{std}(g^y)$	3.8455	3.8522	+0.0067
$\mathrm{acf}(y\text{-cycle})$	0.7721	0.8354	+0.0633
$\mathrm{acf}(g^i)$	0.1335	-0.0792	-0.2126

Synthetic Data Generation and Pretraining

Why Pretrain?

- Embed RBC structural dynamics as a prior over network weights
- Subsequent fine-tuning updates the prior, preserving:
  - Structural dynamics
  - AR(1) persistence of TFP and investment shocks
- Addresses small-sample problem
  - Synthetic pretraining grounds the network before it sees sparse real macro data

Dataset

- Uniform distribution draw centered on calibrated parameter values
  - Range derived from SMM standard errors

- 50,000 RBC simulations × 250 quarters → ~12.5M consecutive training pairs
  - s_t, s_{t+1}
- Input: 4 state variables + 15 noise slots = 19 dimensions

- 15 slots filled with uniform noise from empirical ranges of candidate real-world features
  - Credit Spreads
  - Policy Rates
  - Inflation
  - Unemployment
  - Yield/GDP Proxies
- Target: $\begin{pmatrix} k_{t+1} \\ z_{t+1} \\ \xi_{t+1} \end{pmatrix}$

Architecture

- Residual MLP, ~2.1M trainable parameters
- 19-dim input → projection to hidden = 512 (Linear → LayerNorm → GELU)
- 4 × residual blocks at 512:
  - Linear → LayerNorm → GELU → Dropout(0.1) → Linear → LayerNorm
  - Additive skip connection, then GELU
- 512 → 3 linear output
- LayerNorm + skip connections stabilize deep optimisation; dropout regularizes

Economics-Informed Loss

- Plain MSE ignores structural consistency
  - Impose a two-term loss

- Reconstruct consumption, output, investment from predicted states via
  - Consumptions function
  - Model identities

$$L_{\text{controls}} = E\left(\left|\widehat{c_{\text{pred}}} - \widehat{c_{t+1}}\right| + \left|\widehat{i_{\text{pred}}} - \widehat{i_{t+1}}\right|\right)$$

$$L_{dynamics} = E\left|\hat{k}_{pred} - \left((1-\delta)\hat{k}_t + \delta\hat{i}_t + \delta\hat{\xi}_t\right)\right|$$

$$\text{Total Loss} = L_{\text{controls}} + L_{\text{dynamics}}$$

- Forces predictions to be
  - Pointwise accurate
  - Structurally consistent with consumption function and model dynamics

Training

Epoch	Train	Val
1	0.147	0.065
5	0.063	0.063
20	0.063	0.063
50	0.0625	0.0625

- 90/10 train/val split, fixed seed
- AdamW, lr = 10^{-3}, weight decay = 10^{-4}
- Batch 2,048, cosine annealing over 50 epochs, single GPU
- Val loss trajectory: 0.147 → 0.063 (first 5 epochs) → 0.0625 at epoch 50

Diagnostics

Target	MAE
K_{t+1} (capital)	2.08 * 10^-4
Z_{t+1} (TFP shock)	1.65 * 10^-3
Ξ_{t+1} (investment shock)	1.63*10^-2

- Four structural checks all pass:
  - MAE — capital fully deterministic; shock MAE reflects unforecastable innovations (expected)
  - Autocorrelation — 200-quarter rollout recovers calibrated ρ_z, ρ_ξ
  - IRFs — trace analytical P-matrix responses to 1σ TFP shock
  - Unconditional variance — matches $\sigma^2/(1-\rho^2)$
- Prior saved as W*, passed to Stage 1 fine-tune

Fine Tuning Stage 1: Inter-Country Macro Data

Why Fine-Tune?

Pretrained Network Knows RBC Dynamics but has Never Seen Real Economies

- Real data has frictions, measurement error etc.

Goal: Update the Structural Prior by Training on Several Countries

- Network learns general dynamics
- Can then be adjusted to specific countries or regimes

Data

Panel Covers: Canada, Denmark, Finland, Japan, the Netherlands, Norway, Sweden, Switzerland, the United Kingdom, and the United States

Quarterly Frequency Between 1980Q1 and 2019Q4

Features

4 State Variables from PWT11.0

- HP Filtered
- Investment Shock Backed Out as Consumption Function Residual

$$\widehat{\xi_t} = \frac{\hat{c}_t - a_k \hat{k}_t - a_z \hat{z}_t}{a_\xi}$$

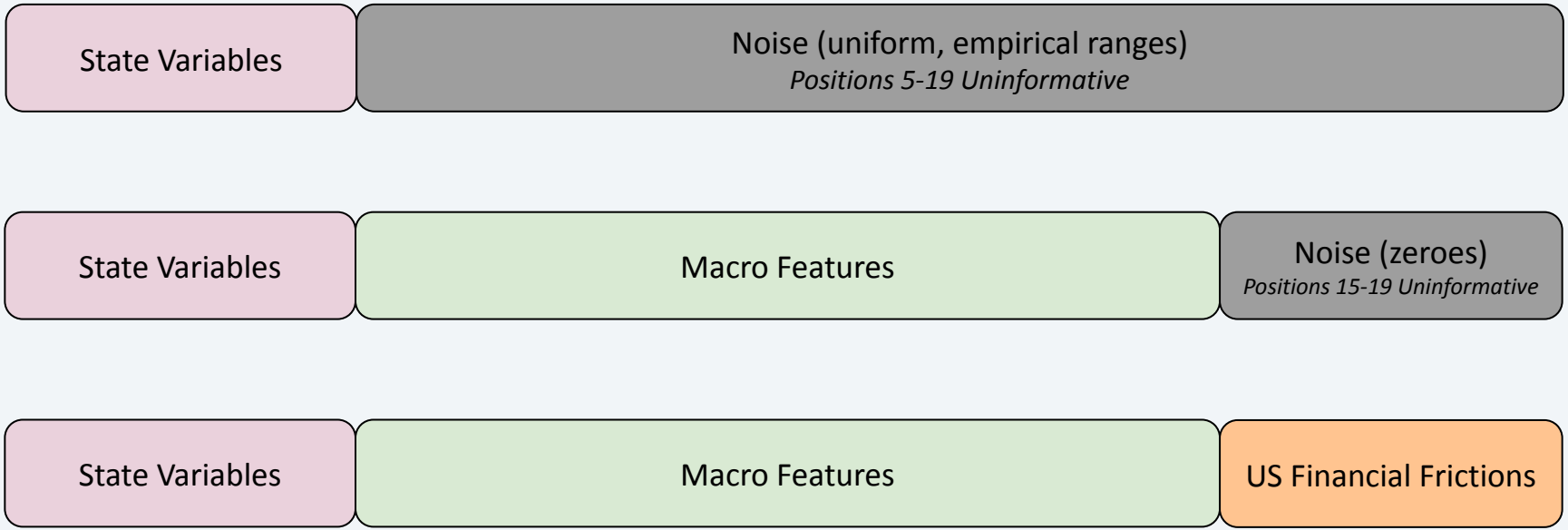
Macroeconomic Features from OECD

- GDP Growth, ULC Growth, CPI Inflation, Unemployment Rate, Yield Spread
- Levels and 8-Quarter Rolling Volatilities
- Seasonally Adjusted

5 Slots Filled with Random Noise for Stage 2 Fine Tuning

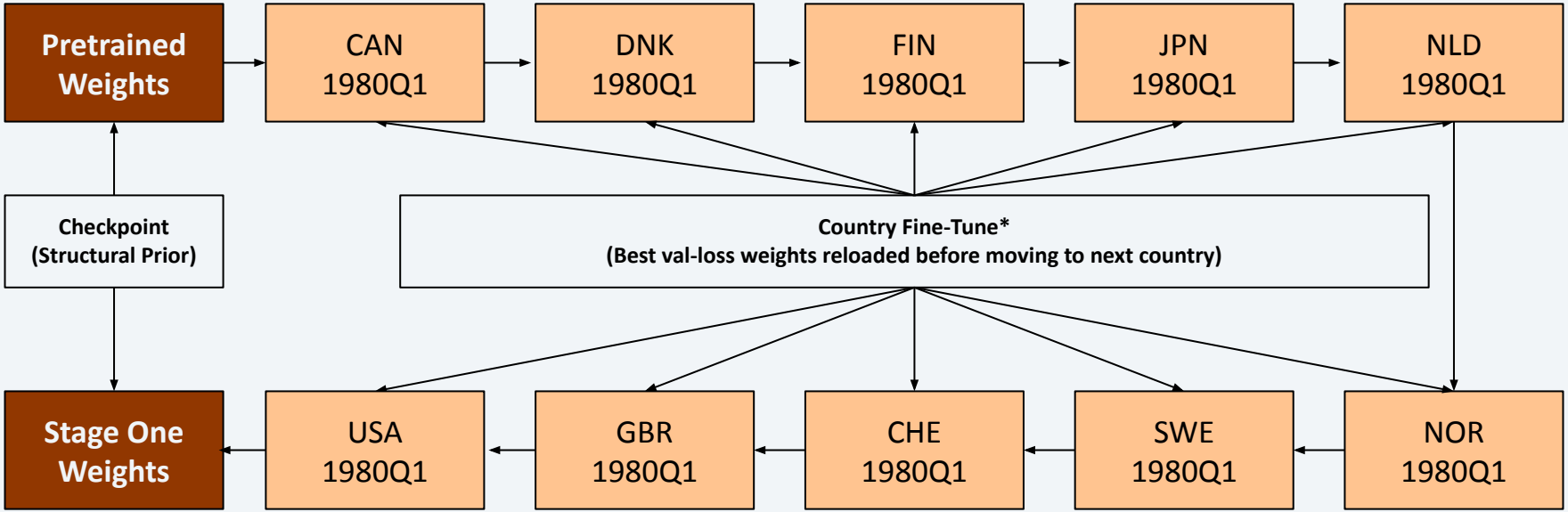
4 State Variables + 10 Macro Features + 5 Noise = 19 Total Features

From Noise to Signal: Replacing the Pad Slots



- **Key takeaway:** No architectural change required at any stage; 19-dim input was reserved from network initialization

Phase One Fine-Tuning Procedure: Sequential Curriculum Learning



*Optimizer reset per-country

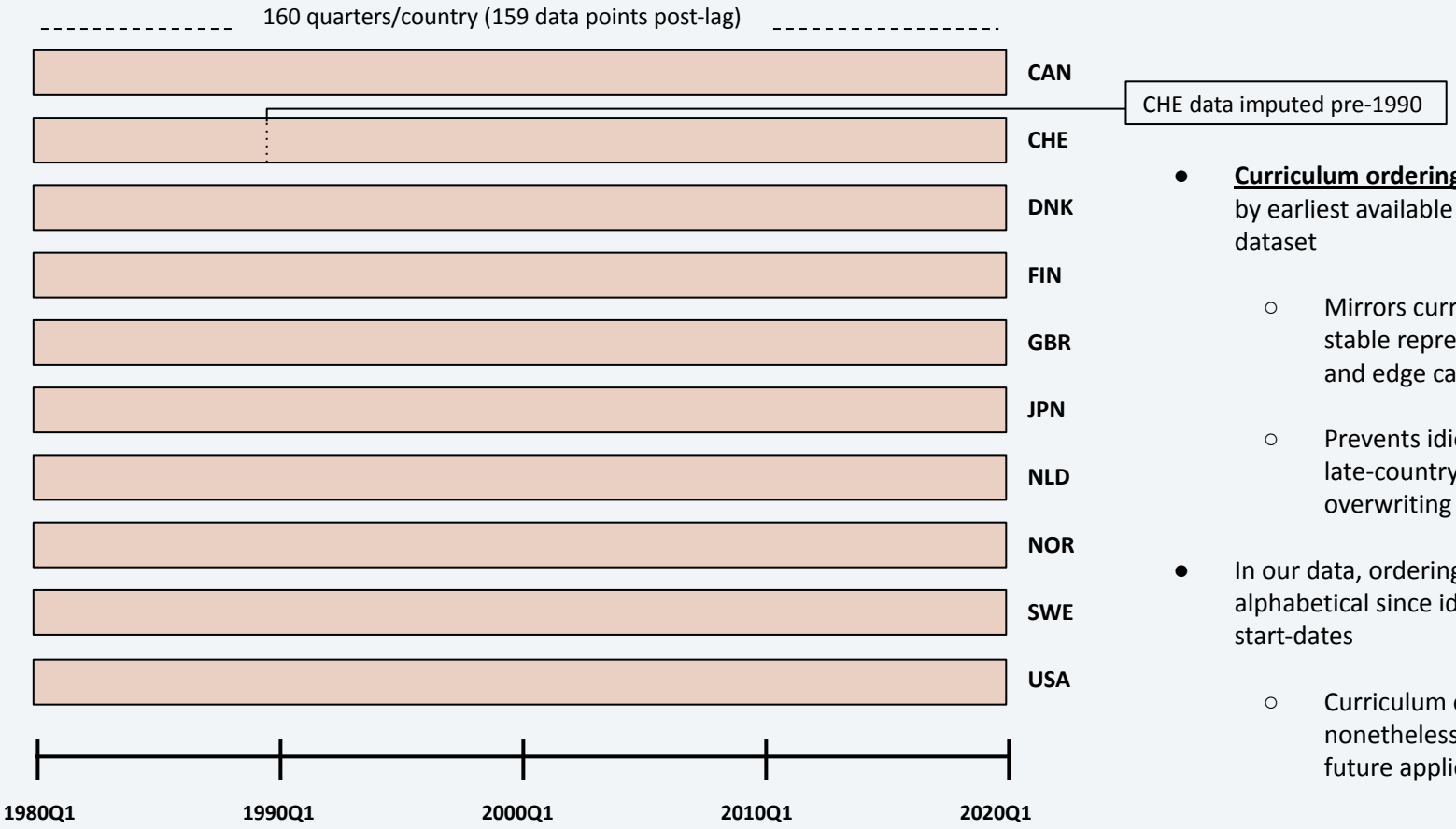
Phase One Fine-Tuning Procedure: Sequential Curriculum Learning

Parameter	Value	Rationale
Learning Rate	10^{-4}	For a conservative update of the structural RBC prior
Optimizer	AdamW	Weight decay regularization*
Batch Size	32	Appropriate for small per-country samples
Max. Epochs	50	With early stopping (patience = 10)
Gradient Clip	1.0	For stability with small samples
Train/Val. Split	80/20	No look-ahead leakage

*Weight decay regularization prevents overfitting by penalizing large weights through a sum of squared weights penalty term added to the loss function

<u>After each country, the **best** val-loss weights are reloaded before beginning the next country (not final epoch weights)</u>

Curriculum Ordering



- **Curriculum ordering:** Countries ordered by earliest available period in the dataset
  - Mirrors curriculum learning; stable representations built first and edge cases later
  - Prevents idiosyncratic late-country dynamics from overwriting structural prior
- In our data, ordering resolves to alphabetical since identical data start-dates
  - Curriculum ordering mechanism nonetheless part of design for future applications

Stage 1 Outcome: What has the Network Learned?

Variable	Pretrained MAE	FT Phase 1 MAE	Improvement
$\hat{k}$ (capital)	0.04307	0.00412	**~90.43%**
$\hat{z}$ (TFP shock)	0.12515	0.00443	**~96.46%**
$\hat{\xi}$ (investment shock)	0.09819	0.09251	**~5.78%**

- Stage 1 of fine-tuning dramatically corrects $\hat{k}$ and $\hat{z}$ but leaves $\hat{\xi}$ nearly unchanged - this is a manifestation of a key motivation for Stage 2
  - Investment shock improvement requires financial friction data (what Stage 2 addresses)

Fine-Tuning Stage 2

Fine Tuning Stage 2: US Financial Frictions

Second Fine Tune

- Phase 1 Taught Network General Macro Dynamics
- Phase 2 Introduces Country-Specific Financial Market Conditions
- Providing Network with Frictions Not Observed from Model
- Hypothesize That This Aligns Structural RBC Prior with Frictional Country-Specific Dynamics

Data

Quarterly Frequency Between 1980Q1 and 2019Q4, Not Seasonally Adjusted

Feature	Source	Friction Addressed
BAA-AAA Spread	FRED	Credit Risk, Borrowing Constraints
Fed Funds Rate	FRED	Monetary Policy, Risk Free Rate
log(VIX)	FRED	Uncertainty, Risk
Term Spread (GS10-FFR)	FRED	Yield Curve, Growth Expectations
log(S&P 500 Return)	Yahoo Finance	Equity Risk Premium

Training Procedure: Conservative Recalibration

Parameter	Stage 1	Stage 2	Reason for Change
Learning Rate	10^{-4}	10^{-5}	Already adapted to real data
Batch Size	32	8	Only 95 training samples
Max Epochs	50	100	Noisier gradient signal
Early Stop Patience	10	20	More time to distinguish signal from noise
Train Samples	~127/country	~95	1990Q1–2019Q4 US only
Val Samples	~32/country	~24	Most recent 20% chronological

- Stage 2 fine-tuning is a targeted recalibration of 5 input channels, <u>not</u> a general retraining
  - Network <u>begins</u> already adapted

Evaluation

Overview

5 Evaluation Dimensions

- Structural Validity (IRFs)
- One-Step Forecast
- Multi-Step Rollout
- Event Studies
- Out-of-Sample Test

Benchmarks

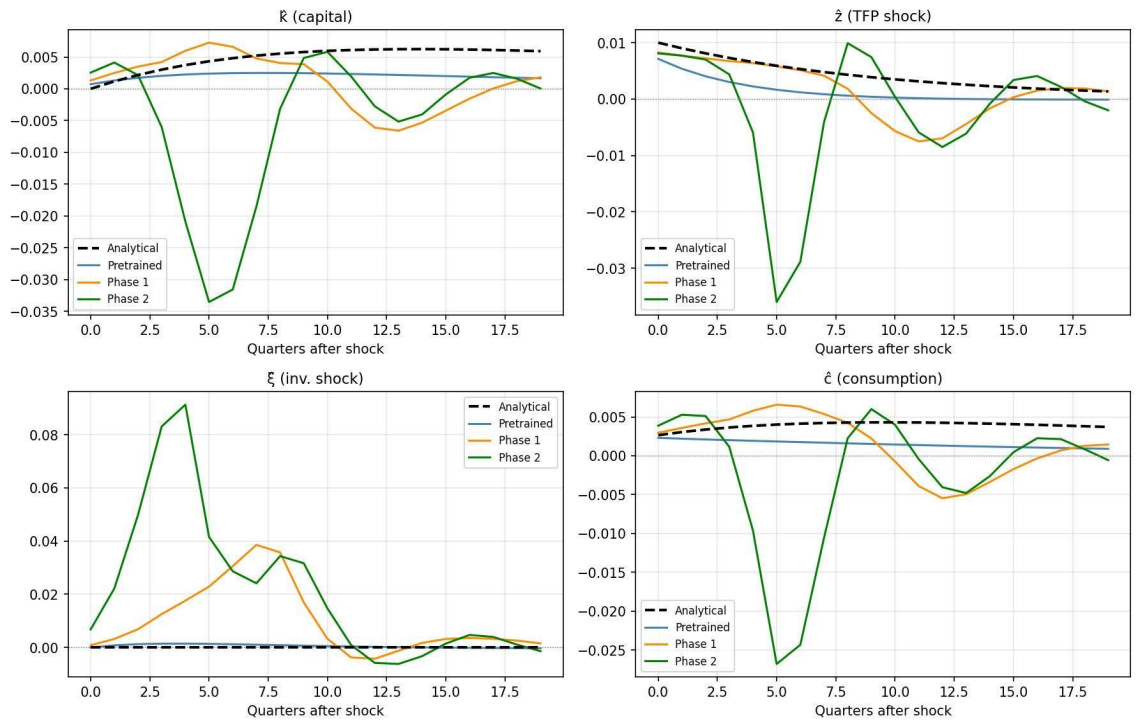
- AR(1)
- VAR(2)
- BVAR

Targets

- Capital Stock
- TFP Shock
- Investment Shock

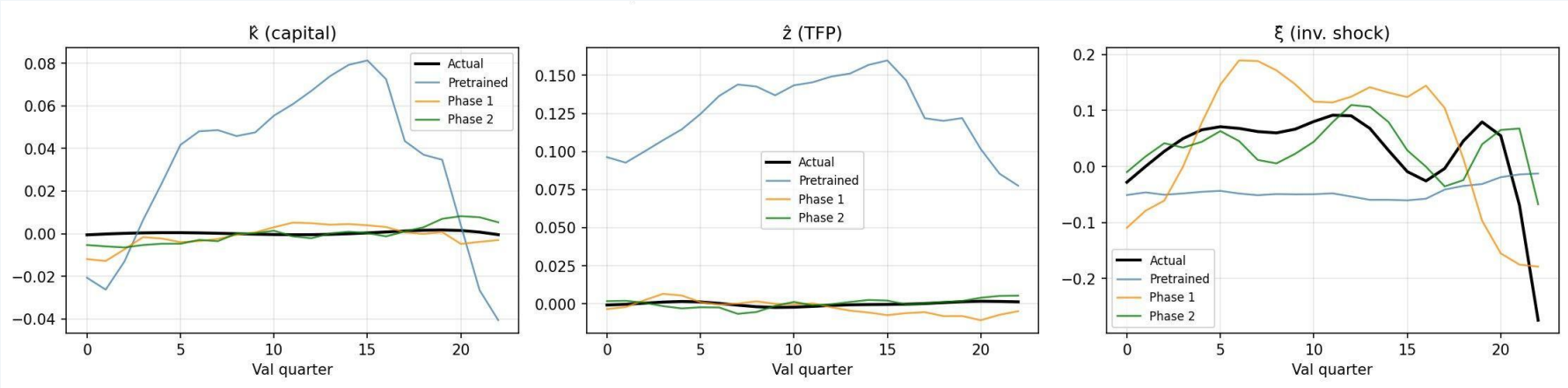
Network Stages Compared Throughout: Pretrained vs. Phase 1 vs. Phase 2

Structural Validity: Impulse Response



- Pretrained Broadly Consistent with Analytical.
- Fine-Tuning Introduces Deviations Given Increased Conditioning Information.
- All Variables Mean Revert, so Structural Coherence is Preserved.

One-Step Forecast

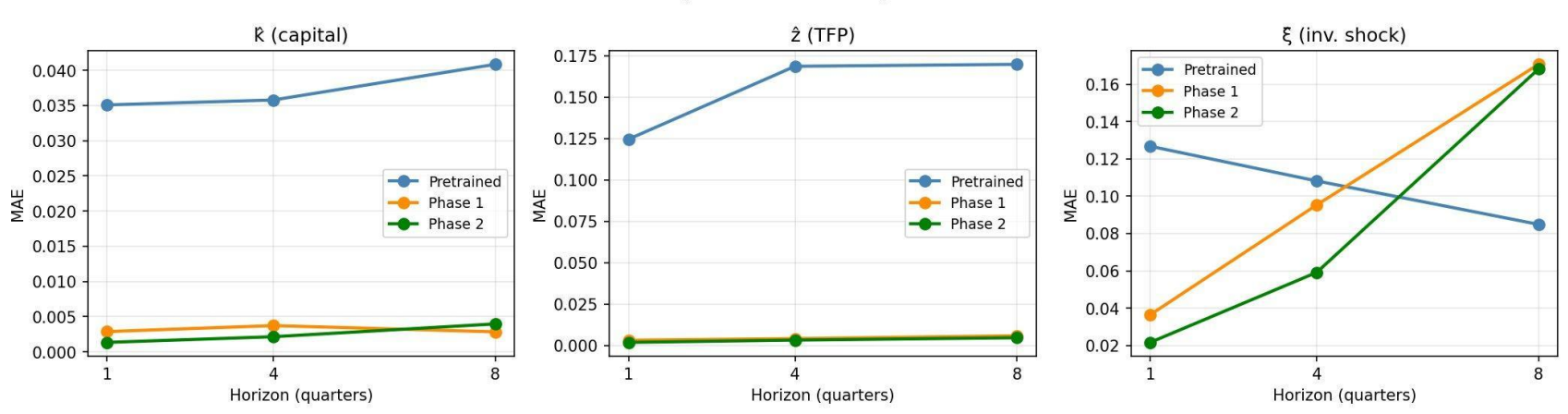


- Pretrained Network is Not Competitive.
- Phase 1 and Phase 2 Yields Large Improvement Over Pretrained Network.
- Phase 2 Adds Significant Value for $\widehat{\xi}_t$ over Phase 1.

One-Step Forecast: Metrics Table

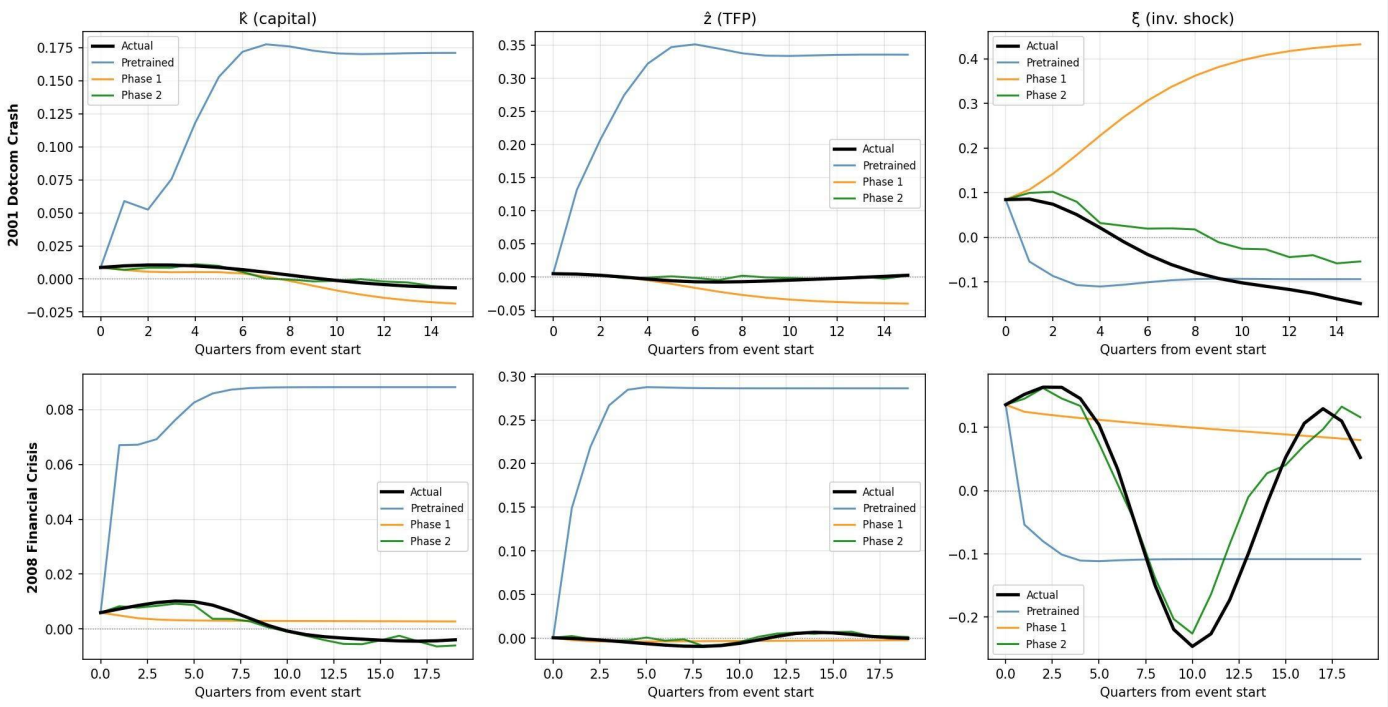
Variable	Model	MAE	RMSE	R^2
$\hat{k}$ (capital)	AR(1)	0.00030	0.00039	+0.706
	VAR(2)	0.00015	0.00021	+0.930
	BVAR	0.00032	0.00038	+0.638
	Pretrained	0.04307	0.04841	−4980
	Phase 1	0.00412	0.00513	−55
	Phase 2	**0.00054**	**0.00061**	**−0.166**
$\hat{z}$ (TFP)	AR(1)	0.00043	0.00054	+0.807
	VAR(2)	0.00021	0.00028	+0.964
	BVAR	0.00057	0.00068	+0.731
	Pretrained	0.12515	0.12754	−10506
	Phase 1	0.00443	0.00545	−18
	Phase 2	**0.00089**	**0.00101**	**+0.337**
$\hat{\xi}$ (inv. shock)	AR(1)	0.03217	0.05418	+0.503
	VAR(2)	0.01888	0.02341	+0.847
	BVAR	0.03303	0.04102	+0.518
	Pretrained	0.09819	0.10991	−1.06
	Phase 1	0.09251	0.10485	−0.87
	Phase 2	**0.04281**	**0.06148**	**+0.357**

Multi-Step Rollout



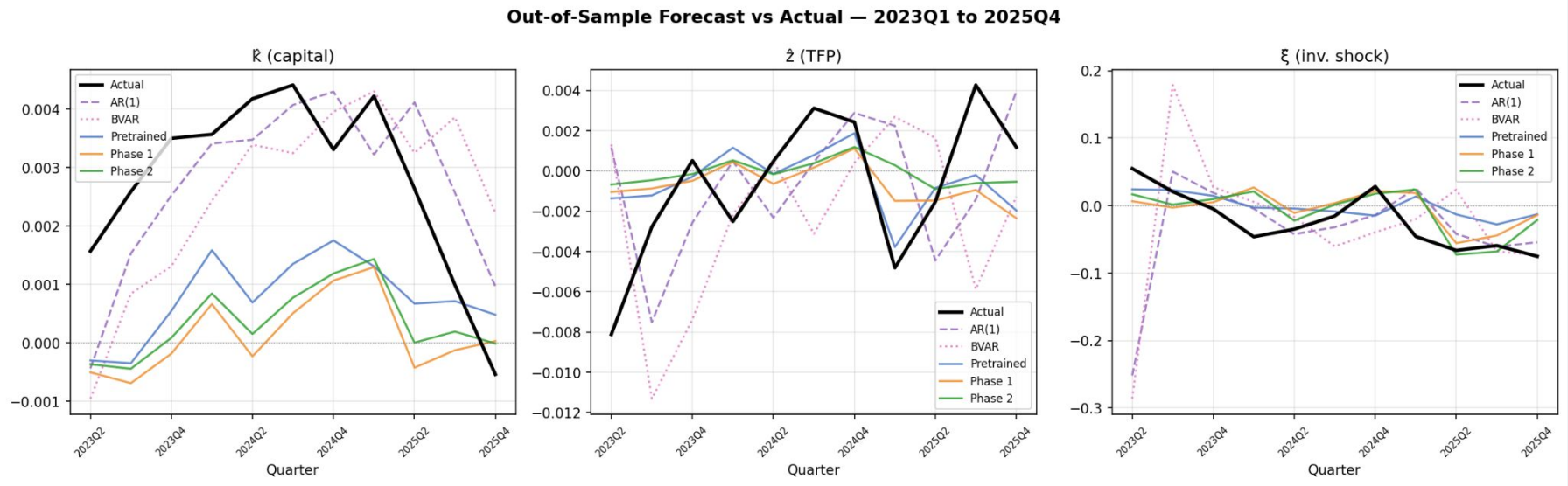
- Phase 1 and Phase 2 Yield Large Improvement Over Pretrained for Capital and TFP.
- MAE's Worsen With Time for Investment Shock.
- Structural Prior Prevents Runaway Error Accumulation.

Event Studies: 2001 and 2008



- Improvement Concentrated When BAA-AAA and VIX Spike.
- 2001 Less Successful (Equity Correction, not Credit Tightening).

Out-of-Sample Test: 2023Q1 to 2025Q4



- Aggressive Tightening Cycle (Post-COVID).
- BVAR and AR(1) Win on Capital Stock, Too Smooth.
- All Networks Beat AR(1) and BVAR on TFP and Investment Shocks.
- Phase 2 Best MAE and R^2 on Investment Shock.
- VAR(2) Completely Irrelevant, Immense Standard Errors.

Out-of-Sample Test: 2023Q1 to 2025Q4 Metrics

Variable	Model	MAE	RMSE	R^2
$\hat{k}$	AR(1)	0.00107	0.00119	+0.344
	BVAR	0.00150	0.00176	−0.432
	Pretrained	0.00219	0.00238	−1.61
	Phase 1	0.00274	0.00296	−3.04
	Phase 2	**0.00251**	**0.00272**	**−2.42**
$\hat{z}$	AR(1)	0.00404	0.00468	−0.77
	BVAR	0.00525	0.00634	−2.263
	Pretrained	0.00233	0.00301	+0.265
	Phase 1	0.00277	0.00338	+0.073
	Phase 2	**0.00276**	**0.00348**	**+0.020**
$\hat{\xi}$	AR(1)	0.05339	0.09800	−4.84
	BVAR	0.07638	0.12094	−7.889
	Pretrained	0.03476	0.03962	+0.046
	Phase 1	0.03237	0.03995	+0.030
	Phase 2	**0.02890**	**0.03687**	**+0.174**

Findings

01 Fine-Tuning is Necessary

Pretrained Network Alone is Not Competitive. Each Fine-Tuning stage reduces MAE and improves predictive capabilities.

02 Network Adds Value

Financial Conditions Inform Investment Shock Best. This can be adjusted with different feature augmentation.

03 Structural Prior Gives Regime Robustness

Flat Multi-Step Errors, Correct 2008 Dynamics, VAR(2) Explodes Beyond 1 Quarter, OOS AR(1) and BVAR Beat, Mean Reversion, Advantage Highest When Stress is Largest, Structural Validity Preserved.

Conclusion

This is a Valid Framework →

Three-stage pipeline: pretrain on theory, fine-tune on macro, fine-tune on finance. Produces a network that is structurally informed and empirically adaptive. Improvement is monotone across all stages.

Strong Baseline →

Conservative design: standard MLP, two-shock linearized RBC, limited features. Richer structural model, transformer architecture, and larger feature set can only improve from here.

Structural Theory and ML are Complements →

The structural prior provides theoretical grounding and alleviates the small-sample problem. Without fine-tuning, the prior alone is uncompetitive. Without the prior, fine-tuning overfits, so both are necessary.

Foundation for Broader Research →

Pipeline is general and accommodates richer models, larger datasets, and more sophisticated architectures.

Discussion

Questions, Feedback, Suggestions

Thank you!